

Lecture 28 : Private-key cryptography (Part II)

Introduction

In the digital age, the security of information is paramount, whether it's personal emails, financial transactions, or sensitive corporate data. Private-key cryptography, also known as symmetric-key cryptography, plays a foundational role in achieving this security by enabling confidential communication between parties who share a secret key. This lecture delves into the practical applications of private-key cryptography, exploring algorithms that have been, and continue to be, instrumental in safeguarding our digital interactions.

We will revisit the evolution of these algorithms, starting with the Data Encryption Standard (DES), understanding its strengths and eventual limitations. This will lead us to its successor, Triple DES, an interim solution designed to bolster security. Finally, we will examine the Advanced Encryption Standard (AES), the current gold standard for symmetric encryption, which underpins much of modern secure communication. Understanding these algorithms is crucial for anyone involved in cybersecurity, as they form the bedrock upon which secure systems are built, ensuring data integrity and confidentiality in an increasingly interconnected world.

Core Concepts

Term	Definition	Significance
Data Encryption Standard (DES)	An older block cipher algorithm, proposed in the 1980s, that encrypts 64-bit blocks of data using a 56-bit key.	Historically significant as a widely adopted standard, but its short key length made it vulnerable to modern brute-force attacks.
Triple DES (3DES)	An extension of DES that applies the DES algorithm three times with three different keys (or two keys in some modes) to enhance security.	Provided increased security over single DES by effectively lengthening the key, serving as an interim solution before AES.

Term	Definition	Significance
Advanced Encryption Standard (AES)	The current widely used symmetric-key encryption standard, also known as Rijndael, offering block sizes of 128 bits and key sizes of 128, 192, or 256 bits.	The most secure and efficient symmetric algorithm today, resistant to known attacks and widely implemented in hardware and software.
Block Cipher	An encryption algorithm that processes plaintext in fixed-size blocks, producing ciphertext blocks of the same size.	Fundamental to DES, 3DES, and AES, allowing for efficient encryption of large data streams by breaking them into manageable units.
Key Size	The length of the cryptographic key, measured in bits, which determines the number of possible keys and thus the difficulty of a brute-force attack.	A larger key size significantly increases the computational effort required for an attacker to guess the key, enhancing security.
Brute Force Attack	An attack method where an adversary systematically tries every possible key until the correct one is found, decrypting the ciphertext.	The primary threat against algorithms with insufficient key sizes, highlighting the importance of robust key lengths.
Feistel Structure	A general design principle used in the construction of many block ciphers, including DES, where encryption and decryption operations are very similar.	Allows for simpler hardware and software implementations as the same core logic can be reused for both encryption and decryption.
Round Keys	Sub-keys derived from the main secret key, which are used in each iterative round of a block cipher's encryption process.	Introduce complexity and diffusion into the encryption process, ensuring that each round's transformation is unique and dependent on the key.

Term	Definition	Significance
Rijndael Cryptosystem	The original name of the algorithm designed by Joan Daemen and Vincent Rijmen, which was selected by NIST to become the Advanced Encryption Standard.	Represents the innovative design that led to AES, showcasing flexibility in block and key sizes before NIST standardized the block size.
NIST	The National Institute of Standards and Technology, a U.S. agency responsible for developing and standardizing cryptographic algorithms.	Plays a critical role in establishing and promoting secure cryptographic standards for government and industry use worldwide.

Detailed Analysis

1. Data Encryption Standard (DES)

The Data Encryption Standard (DES), also sometimes referred to as the Data Encryption Algorithm (DEA), was a pioneering symmetric-key block cipher introduced in the 1980s. It operated on 64-bit blocks of plaintext, transforming them into 64-bit ciphertext blocks. A defining characteristic of DES was its 56-bit key size. For plaintexts longer than 64 bits, the data would be segmented into 64-bit blocks, with the last block potentially padded to reach the required length.

The internal structure of DES is quite intricate, involving a series of transformations over 16 rounds.

- **Initial Permutation:** The 64-bit plaintext first undergoes an initial transposition, or "jumbling," of its bits. This is a fixed permutation that rearranges the input bits.
- **16 Rounds of Processing:** The permuted 64-bit block then proceeds through 16 identical rounds. Each round involves complex calculations, including substitutions and permutations, heavily influenced by a unique "round key."
- **Feistel Structure:** A core design principle within each round is the Feistel structure. The 64-bit block is divided into two 32-bit halves, a left part (L) and a right part (R). In each round i , the new left part L_i is simply the previous right part R_{i-1} . The new right part R_i is computed by taking the previous left part L_{i-1} and performing a bit-by-bit exclusive-OR operation with the output of a complex, non-linear function. This function takes R_{i-1} and the current round key K_i as inputs. This structure ensures that encryption and decryption can use very similar operations, simply by reversing the order of round keys.

- **Round Key Generation:** The 56-bit master key is not used directly in each round. Instead, a key schedule algorithm generates 16 distinct 48-bit round keys. This process involves permutations and left circular shifts of the main key, ensuring that each round uses a different sub-key.
- **32-bit Swap:** After the 16 rounds, the left and right 32-bit halves are swapped.
- **Final Permutation:** Finally, a reverse permutation (the inverse of the initial permutation) is applied to the 64-bit block to produce the final ciphertext.

Despite its widespread use for a significant period, the primary concern with DES was its relatively short 56-bit key length. In the context of modern computing power, a 2^{56} key space is no longer considered secure, making it vulnerable to brute-force attacks by fast computers. This limitation necessitated the development of more robust encryption standards.

2. Triple DES (3DES)

To address the security concerns arising from DES's short key length without completely abandoning the existing DES infrastructure, Triple DES (3DES) was introduced. As its name suggests, 3DES involves applying the DES algorithm three times for each encryption operation. This approach significantly increases the effective key length and, consequently, the security.

- **Operation:** For every 64-bit plaintext block, 3DES performs three sequential DES operations using three distinct 56-bit keys (K1, K2, K3). The most common mode of operation is Encrypt-Decrypt-Encrypt (EDE):
 - The plaintext is first encrypted with K1.
 - The result is then decrypted with K2.
 - Finally, this intermediate result is encrypted again with K3 to produce the final ciphertext.
- **Decryption Process:** Decryption follows the reverse sequence: Decrypt with K3, then Encrypt with K2, then Decrypt with K1. This sequence leverages the property that applying encryption followed by decryption (or vice-versa) with the same key cancels out the operation, effectively reversing the process.
- **Effective Key Size:** By using three 56-bit keys, 3DES achieves an effective key size of 168 bits ($3 * 56 = 168$). This dramatically increases the resistance to brute-force attacks compared to single DES, making it suitable for critical applications for a longer period.
- **Drawbacks:** The main disadvantage of 3DES is its performance. Executing the DES algorithm three times for every block makes it three times slower than single DES. This overhead became a significant concern as data

volumes grew and faster encryption was required. Despite this, 3DES was widely adopted and used in many applications, including financial systems like ATM machines, for a considerable time. However, the performance penalty and the eventual need for even stronger, more efficient algorithms paved the way for a new standard.

3. Advanced Encryption Standard (AES)

The limitations of DES and the performance overhead of 3DES prompted the National Institute of Standards and Technology (NIST) to initiate a competition in 1997 for a new encryption standard. The goal was to find a successor that was more secure, efficient, and flexible. Out of 15 candidate algorithms, the Rijndael cryptosystem, designed by Joan Daemen and Vincent Rijmen, was selected in 2001 and became the Advanced Encryption Standard (AES). Today, AES is the most widely used symmetric-key algorithm globally.

- **Key Features:**

- **Block Length:** While the original Rijndael proposal allowed for variable block lengths (128, 192, or 256 bits), NIST standardized AES to use a fixed block length of **128 bits**.
- **Key Length:** AES offers flexibility in key sizes, supporting **128, 192, or 256 bits**. The choice of key length depends on the desired level of security, with larger keys providing higher security.
- **Implementation Efficiency:** AES was designed for efficient implementation in both hardware and software, a crucial factor for its widespread adoption.
- **Security:** It has proven highly resistant to all known attacks, making it a robust and safe choice for modern cryptographic applications.
- **Internal Structure and Rounds:** Unlike DES, AES does not use a Feistel structure. Instead, it processes the entire 128-bit data block (represented as a 4x4 matrix of bytes, where each byte is 8 bits, totaling 16 bytes or 128 bits) through a series of rounds. The number of rounds varies based on the key size:
 - 128-bit key: 10 rounds
 - 192-bit key: 12 rounds
 - 256-bit key: 14 rounds
- **Round Operations:** Each round (except the first and last) consists of four distinct, sequential transformations:

- **SubBytes:** This is a non-linear substitution step where each byte in the state matrix is replaced with another byte according to a substitution box (S-box). This provides confusion, making the relationship between ciphertext and key highly complex.
- **ShiftRows:** This is a transposition step where the rows of the state matrix are cyclically shifted by different offsets. This provides diffusion, spreading the influence of individual bits across the entire block.
- **MixColumns:** This operation combines the bytes within each column of the state matrix using a mathematical operation over a finite field. This further enhances diffusion by mixing the bits across columns.
- **AddRoundKey:** In this step, the current round key (derived from the main key by a key expansion module) is bit-wise XORed with the state matrix. This introduces the key material into the data transformation.
- **Key Expansion:** Similar to DES, AES employs a key expansion module that takes the initial 128, 192, or 256-bit key and generates all the necessary round keys for the encryption process.
- **First and Last Rounds:** The first round includes an initial AddRoundKey operation before the main round transformations. The last round is slightly different, omitting the MixColumns step.

The sophisticated design of AES, combining substitution, permutation, and mixing operations, along with its variable key lengths, makes it highly secure and efficient, cementing its role as the cornerstone of modern private-key cryptography.

Key Takeaways

- [] DES was an early block cipher with a 64-bit block size and a 56-bit key, which became vulnerable to brute-force attacks due to its short key length.
- [] DES utilizes a Feistel structure, dividing data into left and right halves and applying 16 rounds of transformations with round-specific keys.
- [] Triple DES (3DES) extended DES's security by performing three DES operations (Encrypt-Decrypt-Encrypt) with three different keys, achieving an effective key size of 168 bits.
- [] 3DES provided enhanced security but suffered from a performance penalty, being three times slower than single DES.
- [] AES (Advanced Encryption Standard), based on the Rijndael cryptosystem, is the current standard for symmetric encryption, chosen by NIST for its security and efficiency.
- [] AES uses a fixed 128-bit block size but offers flexible key lengths of 128, 192, or 256 bits, with the number of rounds varying accordingly.

- [] AES rounds involve four core operations: SubBytes (substitution), ShiftRows (row transposition), MixColumns (column mixing), and AddRoundKey (XORing with a round key).
- [] AES is designed for high efficiency in both hardware and software implementations and is considered resistant to all known attacks.
- [] The evolution from DES to 3DES to AES demonstrates the continuous effort in cryptography to develop more secure and efficient algorithms in response to increasing computational power and new attack methods.

Conclusion

The journey from DES to Triple DES and finally to AES illustrates the dynamic nature of cryptography, where algorithms must continuously evolve to stay ahead of ever-advancing computational capabilities and cryptanalytic techniques. These private-key algorithms are not merely theoretical constructs; they are practical tools that form the backbone of secure digital communication. From encrypting emails and securing online transactions to protecting data on hard drives and cloud storage, their integration into various applications provides the essential confidentiality required in our digital lives.

As we conclude our discussion on private-key cryptography, it's clear that the principles of strong key management, robust algorithm design, and continuous evaluation are paramount. While symmetric-key algorithms excel in speed and efficiency for bulk data encryption, they inherently require a secure method for key exchange. This challenge naturally leads us to the next frontier in cryptography: public-key cryptosystems, which will be the focus of our subsequent lecture, exploring how they address the key distribution problem and complement symmetric encryption in building comprehensive security solutions.